SCHOOL MANAGEMENT SYSTEM FOR KYAMATU PRIMARY SCHOOL

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Mwange Musa Muthami

Registration Number: 23/05037

Submission Date: April 2026

Supervisor: Dr. Simon N. Mwendia / Dr. Kevin Mugoye Sindu

A project document submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Information Technology (BIT 3105)

1. INTRODUCTION

1.1 Purpose

This SRS describes the functional and non-functional requirements for the School Management System for Kyamatu Primary School. It serves as the contractual basis between the developer and the school.

1.2 Scope

The system is a web-based application that manages student records, attendance, assessments, reports, fees, timetabling, and communication. It supports Kenya's CBC framework.

1.3 Definitions and Acronyms

Acronym	Definition
SMS	School Management System
CBC	Competency-Based Curriculum
CAT	Continuous Assessment Test
RBAC	Role-Based Access Control
JWT	JSON Web Token
API	Application Programming Interface
ORM	Object-Relational Mapping
SRS	Software Requirements Specification

2. OVERALL DESCRIPTION

2.1 Product Perspective

The SMS is a standalone, web-based system with: Frontend (React 18 SPA via Cloudflare Pages), Backend (RESTful API with Node.js/Express on Render), Database (PostgreSQL 15 with Prisma ORM on Supabase).

2.2 User Classes

Role	Description	Access
Super Admin	Full system access	Full
Admin	School administrator	High
Teacher	Marks attendance, records scores	Medium
Bursar	Manages fees and payments	Medium
Student	Views own dashboard and records	Low
Parent	Views linked student's data	Low

2.3 Operating Environment

- Client: Chrome 90+, Firefox 88+, Edge 90+, Safari 14+
- Server: Node.js 20 runtime
- Database: PostgreSQL 15+
- Minimum Internet: 1 Mbps

3. FUNCTIONAL REQUIREMENTS

3.1 Authentication Module

ID	Requirement	Priority
FR-AUTH-01	Allow user registration with email/password	High
FR-AUTH-02	Authenticate via email/password login	High
FR-AUTH-03	Issue JWT access (15min) and refresh tokens (7days)	High
FR-AUTH-04	Enforce RBAC for all endpoints	High
FR-AUTH-05	Hash passwords with bcrypt (12 rounds)	High

3.2 Student Management

ID	Requirement	Priority
FR-STU-01	Create student profiles with personal details	High
FR-STU-02	Generate unique admission numbers	High
FR-STU-03	Support admission workflow (Pending/Approved/Rejected)	High
FR-STU-04	Link students to guardians	High
FR-STU-05	Assign students to classes	High

3.3 Staff Management

ID	Requirement	Priority
FR-STA-01	Create staff profiles	High

FR-STA-02	Generate unique employee numbers	High
FR-STA-03	Assign teachers to classes and subjects	High

3.4 Academic Structure

ID	Requirement	Priority
FR-ACA-01	Manage academic years	High
FR-ACA-02	Support three terms per year	High
FR-ACA-03	Support CBC grades PP1-Grade 9	High
FR-ACA-04	Manage streams and generate classes	High

3.5 Attendance Tracking

ID	Requirement	Priority
FR-ATT-01	Mark daily attendance per student	High
FR-ATT-02	Support statuses: Present, Absent, Late, Excused	High
FR-ATT-03	Prevent duplicate attendance records	High
FR-ATT-04	Generate attendance reports and statistics	Medium

3.6 Assessment & Grading

ID	Requirement	Priority
FR-ASS-01	Create assessments (CAT,	High

	Mid-term, End-term)	
FR-ASS-02	Record student scores	High
FR-ASS-03	Auto-compute CBC ratings (EE, ME, AE, BE)	High
FR-ASS-04	Support traditional letter grades (A-E)	Medium

3.7 Report Cards

ID	Requirement	Priority
FR-REP-01	Generate termly report cards	High
FR-REP-02	Compute totals, averages, and rankings	High
FR-REP-03	Include teacher/principal comments	Medium

3.8 Fees Management

ID	Requirement	Priority
FR-FEE-01	Define fee structures per grade per term	High
FR-FEE-02	Generate invoices with unique numbers	High
FR-FEE-03	Record payments (Cash, M-Pesa, Bank Transfer)	High
FR-FEE-04	Auto-compute fee balance after payment	High

3.9 Communication

ID	Requirement	Priority

FR-COM-01	Create school-wide announcements	High
FR-COM-02	Target announcements to specific roles	Medium
FR-COM-03	Support direct messaging between users	Medium

3.10 Timetable

ID	Requirement	Priority
FR-TIM-01	Configure timetable slots per class	Medium
FR-TIM-02	Assign subjects and teachers to slots	Medium
FR-TIM-03	Display weekly grid view	Medium

4. NON-FUNCTIONAL REQUIREMENTS

ID	Requirement	Category
NFR-01	API response < 500ms for 95% of requests	Performance
NFR-02	Support 100+ concurrent users	Performance
NFR-03	Passwords hashed with bcrypt (12 rounds)	Security
NFR-04	Helmet middleware for security headers	Security
NFR-05	Rate limiting to prevent brute-force	Security
NFR-06	Responsive design 320px-1920px	Usability
NFR-07	99.5% uptime target	Reliability
NFR-08	Modular feature-based architecture	Maintainability
NFR-09	Environment variables for configuration	Maintainability

5. USE CASES

UC-01: User Login

Actor: Any User

Precondition: User has account

Main Flow:

1. Enter email/password
2. System validates
3. JWT tokens issued
4. Redirect to dashboard

UC-02: Mark Attendance

Actor: Teacher

Precondition: Assigned to class

Main Flow:

1. Select class and date
2. System shows student list
3. Mark each student
4. System saves records

UC-03: Record Scores

Actor: Teacher

Precondition: Assessment created

Main Flow:

1. Select assessment
2. Enter scores
3. System auto-computes grades
4. Scores saved

UC-04: Generate Report

Actor: Admin

Precondition: Assessments scored

Main Flow:

1. Select class and term
2. System computes averages/rankings
3. Report cards generated

UC-05: Record Payment

Actor: Bursar

Precondition: Student has invoice

Main Flow:

1. Select invoice
2. Enter payment details
3. System records and updates balance